

Stock Risk + News

Guessing a Stock's Risk From Price and News — and Whether News
Even Helps

Capstone Project Report

An AI that guesses if a stock will be calm or bumpy next week — plus the surprising answer to 'does adding news actually help?'

RL · RLcapstone.ai

1. Summary

This project guesses whether a stock's next week will be Low, Medium, or High risk — meaning how much it bounces around, **not** whether it goes up or down. It uses 12 clues from the stock's price history plus how positive or negative the news about it was. I mostly built it to answer one question honestly: does the news part actually help?

It got **56% right** out of three choices (random guessing would be 33%), because calm and bumpy weeks tend to come in streaks. But here's the twist: when I removed the news clues completely, it scored **57%** — slightly **better**. So the news didn't help at all.

This is a school project, **not financial advice**. It doesn't tell you to buy or sell anything, and it only guesses how bumpy a stock is, not which way it'll go.

2. Why I built this

Guessing whether a stock goes up or down is basically a coin flip. But guessing how **bumpy** it'll be is more doable, because calm and crazy stretches cluster together. And the whole point of a project 'with news' is to actually check whether the news part is pulling its weight — instead of just assuming it does.

3. The data

I got daily prices for 59 stocks from Yahoo Finance, and news mood (positive, negative, or neutral for each article) from a public 2023 news dataset on Kaggle. From those I built 12 clues per stock per day — recent gains and losses, how bumpy it's been lately, trading volume, big drops, and a daily news-mood score.

No peeking at the future: every clue uses only past info, the thing I'm predicting is the **next** week, and I split training vs. testing **by time** so the test weeks all come after the training weeks.

4. How it works

It's a tiny neural network (about 3,400 numbers) that sorts the 12 clues into Low, Medium, or High. To test whether news helps, I trained the exact same model a second time with the three news clues **removed**, and compared the two.

5. Results

Model (on weeks it never saw)	Right
Price + news	56.3%
Price only	57.2%
Random guessing	33.3%

So the model beats random guessing by a lot — but adding news made it **0.9% worse**, basically no difference. That's actually a normal, honest result: a once-a-day average of news mood is a pretty weak, noisy signal, and the price clues already capture most of what's predictable.

The tempting thing would be to quietly delete the news test and claim 'news helps!' Reporting that it didn't is the difference between a demo and a real experiment.

6. Getting it running

I exported it to a tiny web demo (pick a real stock and week, see the guess vs. what actually happened) and an Android app, both matching the original.

7. Honest limitations

- School project only — not financial advice, and not a recommendation.
- It guesses bumpiness, not direction, and it's only right about half the time.
- Just one year of data and a pretty rough news signal.

8. What's next

- A smarter news signal (actually reading each headline, not just averaging a mood score).
- More years of data covering different kinds of markets.
- Check if news helps more for guessing direction than bumpiness.